

Ontology of Continua as a Bounded Operational Metaontology

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Journal Core Article

This is the narrowed OC Core 1.4 journal-style article surface. The master monograph, evidence atlas, proof appendix, SPOT frontier, and Demonstrator remain companion materials.

Abstract

Ontology of Continua (OC) is presented here as a bounded operational metaontology for describing systems through continua, boundaries, operators, projections, and failure conditions. The contribution of OC Core 1.4 is not a universal empirical law and not a completed formal theory. Its primary contribution is a claim-status discipline: every public claim is routed to a declared support class, evidence surface, proof route, replay route, or explicit non-claim boundary. The accompanying package supplies a 48-claim promoted public projection, a broader Public Science SPOT frontier, proof/status registries, evidence packages, and a public Demonstrator. The article states what the package currently earns, what it does not earn, and how a reviewer can attempt to defeat it.

Primary contribution

OC Core 1.4 contributes a reproducible framework for keeping broad system-language claims answerable to proof, evidence, replay, and failure conditions. The central scientific object is not the phrase “unresolved contradiction” by itself. The object is the controlled route from a proposed description to its claim class, support class, trace, and possible defeat.

Core claims

1. A public OC claim is admissible only when its claim type, support class, source route, and failure boundary are declared.
2. The 48 promoted public claims are closed only at their declared support levels: definition/source-bound or replay-validated.
3. The wider 807-row Public Science SPOT is a frontier map and audit surface, not the promoted theorem set of the article.

What is not claimed

OC Core 1.4 does not claim to be an independently peer-reviewed theory, a universal law of nature, a complete formal system, a replacement for domain science, or an empirical result across all

systems. The package may become stronger only through formal proof, domain data, independent replication, or expert review that is recorded as such.

Methods

The package separates public claims, proof routes, evidence families, source packets, prior-art comparators, Demonstrator data, and release gates. A reviewer can inspect the claim registry, proof closure certificate, Public Science SPOT, evidence package index, and Demonstrator runtime results. The method is fail-closed: a missing route, ambiguous proof status, or unsupported overclaim blocks promotion rather than becoming a rhetorical exception.

Results

The public promoted projection contains 48 claims with zero open promoted claims. The proof certificate records 25 definition/source-bound rows and 23 replay-validated rows. The Public Science SPOT records 807 broader frontier rows, including formal/counterproof and empirical modes, but those rows are not promoted as journal-level conclusions by this article. The Demonstrator supplies an inspectable public graph and runtime route.

Failure conditions

The framework fails locally when a proposed continuum cannot identify what varies, a boundary cannot say what is included or excluded, an operator cannot say what changes, a projection cannot say what survives transfer, or a claim cannot state what would count against it. It fails as a publication package if public text inflates a source-bound definition into a theorem, treats SPOT frontier rows as promoted conclusions, or presents toy numerical examples as evidence.

Discussion

The strongest present status of OC Core 1.4 is a bounded operational metaontology plus reproducible research package. That status is meaningful but limited. It is stronger than an essay because claims are traceable, replay surfaces exist, and overclaim boundaries are explicit. It is weaker than a completed top-tier journal result because independent human peer review and new domain-level validation remain outside the machine-certified package.

Reviewer route

Start with `START_HERE_FOR_REVIEWERS.md`, then read the proof-status glossary, the 48-claim proof certificate, the SPOT frontier boundary, the evidence index, and the Demonstrator route. The monograph is the full corpus; this article is the narrowed claim surface.